Climate Catalyst

GREEN PUBLIC PROCUREMENT: Italy's "Minimum Environmental Criteria"



About this case study

Twenty seven per cent of India's steel demand comes from public construction and infrastructure projects. With steel production expected to triple by 2050, this represents an opportunity to drive sectoral transformation while supporting economic growth.

Green Public Procurement (GPP) isn't just another policy tool, it's how governments worldwide are creating markets for low-carbon steel. In this series, we're exploring how Green Public Procurement is being used and considered by governments across the world to help drive the creation of a market for green steel products. It aims to offer examples of existing and incoming policy which could be relevant in the Indian context.

This is part of a series of case studies which we will release over the course of 2025 and 2026. You can access all of the case studies at https://bit.ly/GPPinIndia

About Climate Catalyst

Climate Catalyst works to compel our political leaders to take action at speed and scale. We do this by galvanising the collective power of businesses, investors and civil society to influence and accelerate policy change. Working behind the scenes we bring together new and diverse stakeholders to identify opportunities with the greatest potential for collaboration, share ideas and learnings and deliver creative campaigns that build power and secure decisive action by governments that reduce greenhouse gas emissions. Our vision is for a just, prosperous world in which global temperature rise is limited to 1.5°C.

Find out more at www.climatecatalyst.org,
on X @climacatalyst
or email info@climatecatalyst.org.

Introduction

Italy's Minimum Environmental Criteria (CAM) was introduced as part of a national effort to use public procurement as a regulatory tool to drive environmental performance in construction and materials markets. It is one of the few procurement frameworks where environmental standards are mandatory and legally binding across public tenders. The gradual yet vigorous approach to embedding environmental standards in the CAM offers important lessons for India's Green Public Procurement framework for low carbon steel. In the last two decades, Italy has successfully integrated standards which cover the entire public tender lifecycle for all construction and renovation projects.

Italy has been encouraging voluntary action from firms. Compliance with the CAM was made mandatory in 2015. Since then, many private firms have voluntarily adopted environmental standards set by this public procurement policy. By working with industry, the Italian government has achieved a balance whereby competition between firms has driven innovation. Where the market has been slower to adopt voluntary standards, the government has encouraged adoption by signalling eventual mandatory requirements. This has been transformational for the Italian steel industry; 84 per cent of steel produced in Italy is made from recycled scrap.

Italy's experience in developing the CAM has some key take-aways. The gradual approach of introducing incremental mandatory criteria for design and construction companies bidding for public contracts has had huge success in breeding a competitive and voluntary drive from industry to improve environmental standards.



About the CAM

When did it come into force? The Minimum Environmental Criteria (*Criteri Ambientali Minimi, CAM*) were first introduced in Italy in 2008 as part of the National Action Plan for Green Public Procurement (PAN GPP). These initially voluntary guidelines were amended in 2015, making the CAM mandatory for all relevant public tenders. The CAM are updated regularly and further integrated into procurement law. The current CAM came into force in 2022 with specific definitions for construction materials and public works. Moreover, the 2023 Public Procurement Code (PPC) sets out in Article 57 that the technical specifications and contractual clauses contained in the CAM must apply for the entire value of the tender. The PPC also includes the CAM in the award criteria for tenders, as is discussed below.

Who does it apply to? The CAM have been gradually integrated into Italy's procurement laws, and now apply to all the stages in public procurement contracts. The criteria must be met by the project designers and the construction companies who carry out the building or renovation works. Many private firms have voluntarily integrated environmental standards in their production to increase their competitiveness when bidding for public tenders.

The CAM have been developed for 18 product categories and 20 procurement categories covering public procurement from interior furniture to building, and vehicles to incontinence aids.

How does it work? The CAM operate through a structured set of requirements embedded in tender documents. As Sabina Nicolella at the *Fondazione ecosistemi* puts it, "Italy produces criteria that reflect the way that public administration carries out procurement. The CAM cover all types of projects and materials which make up procurement."

Each CAM document specifies technical specifications that define minimum environmental performance standards for products and services. For example, in the construction sector, they require the use of steel with a minimum recycled content, depending on the type and quality of steel used and its use in the project. In addition to technical specifications, the CAM include award criteria that allow contracting authorities to give additional points to tenders offering superior environmental performance. This creates a competitive advantage for suppliers that go beyond the minimum requirements. The CAM also contain contract clauses that impose obligations for monitoring and reporting during project execution. These clauses ensure that environmental commitments made during the tendering phase are implemented in practice. This competitive element based on environmental standards means the CAM are part of a bigger story of market-led innovations for green transformation rather than a top-down government mandated approach.

The CAM are not environmental certifications; rather, they are standards. The CAM require compliance of procedures, composition, type, quality, and intended use of the construction component. Only when a product meets these requirements is it considered CAM-compliant.

The first key lesson to draw out from the CAM for India is that embedding low-carbon steel in all aspects of public procurement is essential. The CAM's success stems from its mandatory nature and its integration into the Public Contracts Code. This legal embedding ensures that sustainability requirements are not optional but a core part of procurement law. At the same time, the CAM are designed to encourage active participation in voluntary measures through market-led competition. This means Italy's industry has not remained static in high-carbon steel production. Instead, companies vying for public contracts have helped to spearhead the wide adoption of scrap-based production. For India, clear statutory backing for green procurement standards will be critical to avoid ambiguity and ensure consistent application across ministries and agencies.

Successes of the CAM

The CAM have achieved notable success in transforming Italian public procurement. One of the most significant achievements is their mandatory nature. Unlike many European countries where green procurement remains voluntary, the CAM are compulsory for all relevant tenders in Italy. The National Anti-Corruption Authority (*Autorita Nazionale Anti-Corruzione, ANAC*) has clarified that no derogations are permitted, making the CAM a non-negotiable element of public contracts.

This mandatory application has created a strong incentive for suppliers to adopt sustainable practices. Companies that fail to comply risk exclusion from lucrative public projects, particularly those funded under the National Recovery and Resilience Plan, which saw a construction and renovation boom in Italy post-pandemic. As a result, there has been a rapid increase in demand for certified products and services, and many companies now choose to develop CAM-compliant components on a voluntary basis. Italy now ranks among the top European countries for green public procurement implementation. According to studies by *Fondazione Ecosistem*i, billions of euros in public contracts require CAM compliance, particularly in the construction, infrastructure, and energy sectors.

There has also been a significant increase in the availability of certified recycled materials. Environmental certifications such as Environmental Product Declaration (EPD) ReMade in Italy have grown rapidly. Life cycle thinking has become integrated into design and procurement processes, which represents a cultural shift in how projects are conceived and executed.



However, challenges remain. Small and medium-sized enterprises may struggle with certification and compliance costs, which can be a barrier to participation in public tenders. The supply of certified recycled steel is also finite, and growing demand often outpaces availability, creating market pressure. Monitoring and enforcement also present difficulties, as ensuring compliance across thousands of tenders requires robust oversight mechanisms. A challenge that would be true in the Indian case as well.

Overall, the Italian case demonstrates how leveraging procurement as a market-shaping tool can have enormous benefits for both government and industry. The CAM have created strong demand signals for recycled steel and low-carbon materials, accelerating investment in scrap-based production and certification systems. India can use its large public procurement footprint to normalise the "green premium" by allowing ministries to pay modest premiums for verified low-carbon steel. This early-stage demand can de-risk investments in cleaner technologies and signal confidence to private investors.

Interaction with other policies

The CAM do not operate in isolation. They interact with and reinforce several national and European policies. At the European level, CAM align with Green Public Procurement Guidelines, which encourage member states to integrate sustainability into public purchasing. They also support the objectives of the Circular Economy Action Plan by promoting resource efficiency and recycling. CAM criteria reflect the requirements of the European Taxonomy for Sustainable Activities, which defines what constitutes environmentally sustainable economic activity.

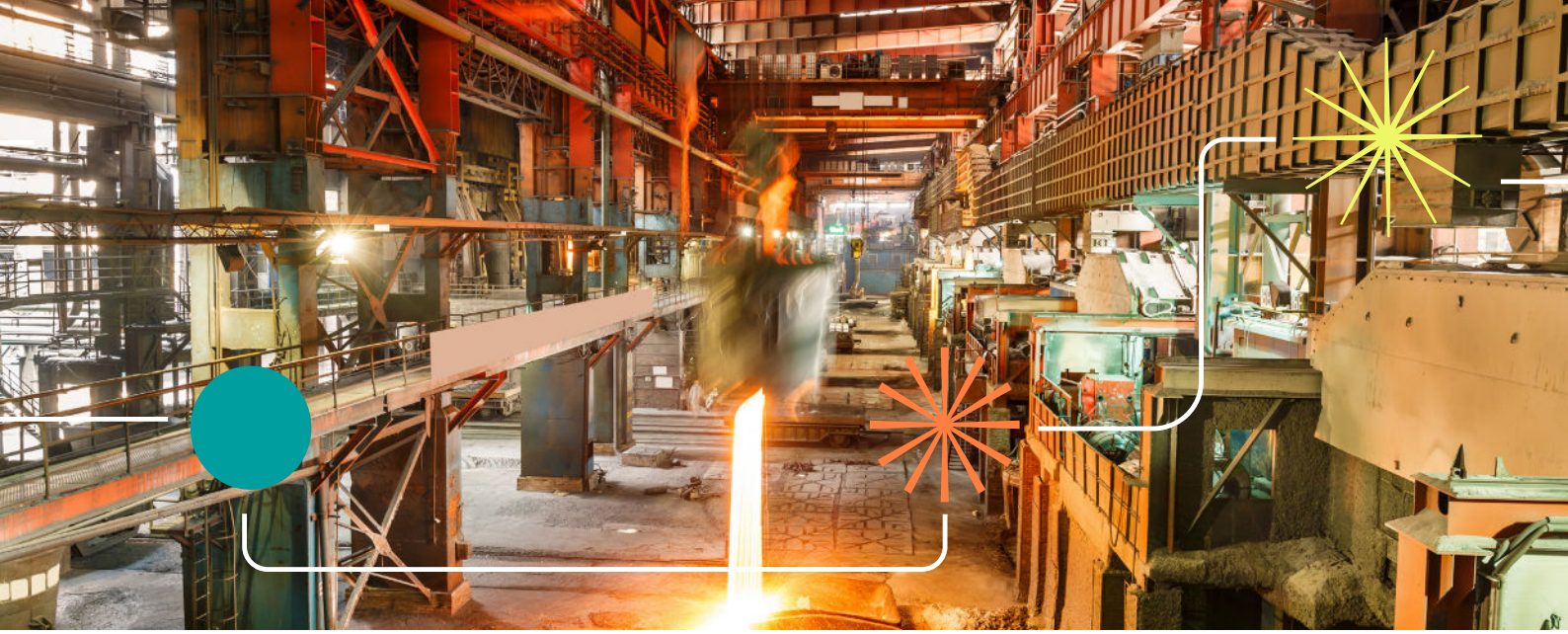
At the national level, CAM are integral to Italy's green transition strategy under the €194-billion-National Recovery and Resilience Plan (PNRR), which is funded by the EU's Next Generation EU fund. Italian green public procurement also features as key points for other national goals, including the National Strategy for Circular economy and the National Plan for the Ecological Transition.

They also reference international standards such as EN 15804 for Environmental Product Declarations and EN 15978 for building life cycle assessment. This synergy ensures CAM are consistent with broader sustainability frameworks, enhancing their legitimacy and effectiveness. By embedding CAM within a network of complementary policies, Italy has created a coherent approach to sustainable procurement that aligns with both domestic priorities and European objectives.

Sabina Nicolella and Dana Vocino at the *Fondazione Ecosistemi* commented that "At the moment, life cycle assessment (LCA) is a premium criterion, but over time, premium criteria become mandatory. This gives time for companies to adapt. Currently, LCA are very detailed because the Ministry of the Environment is considering making LCA mandatory for projects."

There is scope for the Italian Ministry of the Environment to bring in the global warming potential impact as part of LCA for CAM compliance. Such a move would see steel include not just recycled content criteria, but also emissions criteria.

India could consider empowering a single nodal agency to coordinate across ministries, set timelines and publish verified procurement data. This will provide continuity and reduce fragmentation in policy implementation. For Italy, this holistic approach has helped the CAM to remain relevant and targeted on the conditions of market supply and demand. Institutional coordination is therefore key to the success of new "green" steel regulation.



Recycled steel: Requirements

Steel is a critical material in construction and infrastructure, making it a focal point of CAM requirements. The 2022 update introduced stringent recycled content thresholds for steel products. For structural steel produced using the electric arc furnace process, the recycled content must be at least 75 per cent. For alloy steel, the requirement is 60 per cent, while for steel produced through the integrated cycle, the threshold is 12 per cent. For non-structural steel, the requirements are slightly lower but still significant: 65 per cent for electric arc furnace non-alloy steel, 60 per cent for alloy steel, and 12 per cent for integrated cycle steel.

Verification of compliance is rigorous. Self-declaration is no longer accepted. Producers must demonstrate compliance through Environmental Product Declarations of Type III, ReMade in Italy certification, or other accredited third-party certifications. This shift from self-certification to independent verification reflects a broader trend towards transparency and accountability in sustainability reporting.

The implications for producers are profound. They must adjust their supply chains to secure sufficient scrap material and optimise production processes to meet recycled content thresholds. Investment in certification and life cycle assessment documentation is necessary, which adds cost and complexity. However, compliance also offers competitive advantages. Early adopters gain access to public contracts and strengthen their market position. For producers using traditional blast furnace methods, the pressure to innovate is intense, as meeting CAM requirements is challenging within the constraints of the integrated cycle.

CAM also address logistics and carbon footprint. They encourage local sourcing by limiting transport distances to 150 kilometres from production site to construction site. By combining material requirements with transport considerations, CAM promote a holistic approach to sustainability.

Defining clear thresholds for environmental standards is another important lesson for India. Similar specifications relating to recycled steel content in the CAM could be adopted in India, provided this is supported by independent certification. In the CAM, self-declaration is explicitly prohibited. India could adopt a similar approach by setting transparent, performance-based criteria for carbon intensity or recycled content and mandating third-party verification. This will build credibility and prevent greenwashing. Furthermore, striking a balance of ensuring technology neutrality while rewarding performance is highly rewarding. Italy's CAM apply to all steel routes but set differentiated thresholds that reflect technical realities: electric arc furnace producers must meet higher recycled content levels, while integrated cycle producers have lower yet meaningful requirements. India's steel sector is diverse, combining large BF-BOF plants, induction furnaces, and emerging EAF producers. A performance-based approach that rewards actual emissions reductions rather than legacy technology choices will avoid distortions and encourage innovation across all routes.



Where next for the CAM

The CAM are dynamic and will continue to evolve. The current recycling focus of the CAM is likely to expand to include emissions-based criteria. Moreover, life-cycle costing (LCC) has already begun to be integrated for the public procurement of both products and services. Italy's central purchasing body, Consip (Concessionaria Servizi Informativi Pubblici), has developed a unique approach for introducing LCC for selected tenders, with a simplified methodology which is adjusted for each product or service group. For example, the CAM set out thresholds for energy consumption for lamps. As Vocino notes, "at the moment, lifecycle assessment is a premium criterion. The CAM have developed in such a way that over time, premium criteria become mandatory. The CAM introduces premium criteria, then gives time for companies to adapt, and then gradually make these criteria mandatory".

Other future directions include expansion to new sectors such as information and communication technology, energy services, and road maintenance. Digitalisation of compliance processes is likely, with the development of centralised Environmental Product Declaration (EPD) databases and blockchain-based traceability systems. Stricter carbon benchmarks may be introduced to align with European climate targets, which could include carbon intensity limits for materials.

Support for small and medium-sized enterprises will be essential. As Costanza Scano at ECCO points out, "including ETS emissions will not add any significant administrative burdens for large industries applying for tenders. However, for small and medium enterprises, this burden will be significant." Simplified compliance tools and training programmes can help reduce barriers to participation. Enhanced monitoring and enforcement mechanisms will also be necessary to ensure that CAM deliver on their promises. ANAC and regional authorities are expected to play a key role in strengthening oversight.

The CAM will remain a cornerstone of Italy's sustainability strategy. By shaping supply chains and accelerating the transition to a circular economy, they will continue to influence policy, industry, and the broader sustainability agenda.

Italy's CAM demonstrate that procurement can be a powerful lever for industrial decarbonisation when it combines legal enforceability, transparent standards, institutional clarity, and alignment with wider policy objectives. If India adopts these principles, its forthcoming green procurement framework could accelerate the steel sector's transition and position the country as a global leader in credible, market-shaping sustainability policy.



Summary of Lessons for India

The CAM framework demonstrates how public procurement can drive systemic change. By embedding environmental criteria into tenders, Italy has created a powerful incentive for suppliers to innovate and adopt sustainable practices. For steel producers, the CAM requirements on recycled content represent both a challenge and an opportunity. Those who adapt will thrive in a market increasingly defined by green standards. As the CAM evolve, they will continue to influence policy, industry, and the broader sustainability agenda. Italy's experience with the CAM therefore offers several constructive lessons for India as it develops its own Green Public Procurement framework for low-carbon steel.

Several principles stand out. Italy illustrates the importance of embedding low-carbon requirements across the entire procurement system so that sustainability criteria are consistently applied. Clear and measurable thresholds, supported by independent verification, help to build credibility and prevent greenwashing. Procurement can also serve as a powerful market signal that supports early investment in cleaner technologies by reducing uncertainty for producers.

Italy has also shown the benefit of maintaining technology neutrality while rewarding strong performance. In this way, government regulation targets environmental outcomes and emissions reductions rather than technological methods used by companies. All production routes are considered, but requirements are designed in a way that reflects technical realities and encourages continuous improvement. Stable institutional coordination has helped the CAM remain relevant and up to date, while alignment with broader industrial and climate strategies has ensured that procurement works in tandem with other policy tools

Italy's CAM demonstrate that procurement can be a powerful lever for industrial decarbonisation when it combines legal enforceability, transparent standards, institutional clarity, and alignment with wider policy objectives. If India adopts these principles, its forthcoming green procurement framework could accelerate the steel sector's transition and position the country as a global leader in credible, market-shaping sustainability policy.

Interviewees

This case study is based on desk research and interviews with the following low-carbon procurement experts:

- Costanza Scano, Researcher Industrial Transformation at ECCO.
- Sabina Nicolella, CSR, Stakeholder Engagement and European Planning at Fondazione ecosistemi.
- Dana Vocino, GPP, DNSH, Territorial Planning and Sustainability Reporting at Fondazione ecosistemi.

References

* ECCO, *Technical Report: Una strategia per l'acciaio verde opzioni e sfide della decarbonizzazione* (2022),
* Fondazione ecositemi – Forum Compraverde, *Green Public Procurement legislation in Italy*
* OECD, *Harnessing Public Procurement for the Green Transition: Good practices in OECD countries* (2024), OECD Public Governance Reviews.



Sangeeth Selvaraju is a Policy Fellow at the Grantham Research Institute on Environment and Climate Change at the LSE and holds a joint appointment with the Stern team and the Just Transition Finance Lab. His work focuses on hard-to-abate sectors, just transition and transition finance in emerging markets and developing economies.



Mark Nicholls is an independent journalist and editor specialising in sustainable finance and environmental markets. He was co-founder and editorial director of *Environmental Finance* magazine.

Climate Catalyst

W: www.climatecatalyst.org
E: info@climatecatalyst.org
T: @climacatalyst
L: linkedin.com/company/climate-catalyst-ngo

February 2026